
Oliver Krauss

Introduction

ModernCode | MTD.ba VZ SS25

Hagenberg 06.10.2025

HAGENBERG | LINZ | STEYR | WELS



UNIVERSITY
OF APPLIED SCIENCES
UPPER AUSTRIA

Course Instructor

Oliver Krauss

- SE Bachelor & Master in Hagenberg
- Member of research group AIST since 2012
- Lectures since 2015
- oliver.krauss@fh-hagenberg.at
- +43-(0)50804/27195
- Office: FH Hagenberg - FH1.233



What we cover in Modern Code Development

1. Development Environment & Java Basics
2. Variables, Types, and Control Structures
3. Arrays, Strings, and File Operations
4. Methods, Modularity, and Debugging
5. 2D Arrays, Problem Solving Paradigms
6. AI-Assisted Coding and Code Quality

Organizational matters I

Attendance and participation

- Attendance is NOT mandatory
- Handing in the exercises IS mandatory
- Participation (if attending) is mandatory
- Both attendance and participation can only influence your grade positively

Organizational matters II

Teaching Method

- Slide Presentations
- Live coding demos & practical work with your participation
- Hands-on projects using modern development tools
- All Material on Moodle

Organizational matters III

Grading

- Lecture: Written **exam** (90min. ModernCodeVO grade)
- Exercises: **Homework** via Moodle + **Short Tests** (ModernCodeUE grade)
 - **Homework:** 8 assignments, 24 points each
 - **Short Tests:** 3 tests, only best 2 count
 - Handing an assignment in late -> -4 points / day
 - Deadlines will be extended only in case of emergency.
 - The entire group may ask for a deadline extension
 - You can Skip ONE assignment. Assignments only count as submitted if you reach at least 6 points (25%)

Organizational matters IV

The Effort:

- This Class is worth **2.5 ECTS** (x2 -> VO, UE)
- You should spend 62,5h on this class (In Austria 1 ECTS = 25h of work)
- 30 hours in class (if you attend!); 32,5 hours at home
- Please document your hours. It will allow me to align the class better next year.

Modern Code Development- Why it matters

Industry Relevance

- Modern software development practices are essential for today's job market
- Companies expect developers to know version control, testing, and deployment
- Code quality and maintainability are crucial for long-term project success

Learning Outcomes

- Master modern development workflows and tools
- Understand code quality principles and testing strategies
- Learn to write maintainable, scalable code
- Gain practical experience with real-world development practices